

Aerial Cetacean Detection for Autonomous Offshore Monitoring: One Detector across the Ground-Sample-Distance Axis, from Exact Detection to Wide-Area Screening

Gerard Dooly¹ and Petar Trslic¹

¹SurveyLabs Ltd., Ireland , {gerard.dooly, petar.trslic}@surveylabs.ie

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Technical report — work in progress. Data sources, the detector and the reported results are updated as the AEROSUB project progresses.

Abstract

Environmental monitoring of cetaceans around offshore-wind developments is labour-intensive, relying on trained observers and slow manual review of survey imagery. As part of the HORIZON Europe project **AEROSUB** (GA 101189723), SurveyLabs is developing an autonomous aerial monitoring capability in which a long-range vertical take-off and landing (VTOL) fixed-wing UAS detects cetaceans over open water. This report documents the RGB perception component. We first consolidate ten heterogeneous aerial and near-surface sources into a unified, deduplicated dataset of 27,290 images / 55,163 annotations through a reproducible normalisation pipeline that fixes label-format, EXIF-orientation and offline-augmentation inconsistencies, and we describe a Label Studio human-in-the-loop pseudo-labelling loop for growing it. Our organising argument is that a single aerial detector is asked to operate across a wide *ground-sample-distance* (GSD) axis, and that the two ends of that axis are two different tasks: at *high* GSD the animal is resolved to hundreds of pixels and the task is *exact detection* (localise and count), whereas at *low* GSD the animal collapses to a few pixels and the only realistic task is *wide-area screening* — cueing a human expert at high recall for a small false-alarm budget. A six-way controlled study (varying backbone capacity, data composition and hard-negative dose one factor at a time) shows that low-GSD screening recall is governed by data composition and capacity rather than tuning: removing the porpoise source collapses the detector, extra surface data hurts, and — reversing a common assumption — the larger **YOLO11s** backbone lifts screening recall substantially (0.535 vs 0.412 recall at 1 false positive per image) even though it barely moves precision-regime mAP. We then introduce **adaptive SAHI**, a telemetry-driven slicing scheme that ties tile size to GSD so the detector sees animals at a roughly consistent apparent scale. In a single-frame illustration on a real 151-megapixel survey frame, plain whole-frame inference detects *none* of three resolvable dolphins whereas adaptive SAHI recovers all three with zero false positives at 17× fewer tiles than the best fixed slice; a semi-synthetic altitude sweep shows the same policy holding recall comparable to fixed slicing while spending tiles only as geometry demands. Finally we quantify the practical limit of the low-GSD end: on out-of-domain survey frames the detector plateaus near 0.6 recall at 2 false positives per image — useful triage, but a real domain gap that further in-domain pseudo-labelling is designed to close. The model, data pipeline and inference tooling form the basis of AEROSUB deliverable D2.1 and key exploitable result KER9.

Keywords: cetacean detection, aerial imagery, UAS, YOLO, small-object detection, SAHI, ground-sample distance, offshore wind, marine mammal monitoring.

1 Introduction

Offshore wind is expanding rapidly, and with it the regulatory requirement to monitor the impact of construction and operation on marine megafauna. Cetaceans (whales, dolphins and porpoises) are protected species whose presence must be assessed before and during noisy activities such as pile driving. Present practice relies on *marine mammal observers* stationed on vessels or platforms, and on manual review of aerial or vessel survey imagery. This is expensive, weather-limited, spatially restricted, and slow: latency from image capture to a confirmed sighting can be tens of minutes.

Within the HORIZON Europe project **AEROSUB** (Grant Agreement 101189723), Survey-Labs is tasked with on-board *detection and tracking* of cetaceans from RGB aerial imagery. This report documents that work: a robust aerial cetacean detector, the data pipeline that feeds it, and the survey-scale inference machinery around it.

Central argument: one detector, one GSD axis, two tasks. The organising thesis of this report is that open-water detection reliability is governed less by raw model capacity than by *data composition* and *scale-aware inference*, and that these two levers act at opposite ends of a single continuum — the **ground-sample distance** (GSD, metres of water per image pixel) at which an animal is imaged. A survey aircraft does not fly at one scale: altitude, lens and sensor set how many pixels a given animal subtends, and that number spans orders of magnitude across a campaign. We argue that the two ends of this GSD axis are not the same problem measured more or less precisely — they are *different tasks* that deserve different metrics:

- **High-GSD, exact-detection regime.** When the animal is resolved to hundreds of pixels (low altitude, high-resolution sensor, or scale-recovered by tiling), the operational task is to localise and count individual animals. The right metric is precision and mean average precision (mAP) against exact ground-truth boxes. The real anchor for this regime is a 151-megapixel survey frame at $\text{GSD} \approx 2.4 \text{ cm/px}$ (Section 5.5), where scale-aware inference recovers the full pod.
- **Low-GSD, screening regime.** When the animal collapses to a handful of pixels (high altitude, coarse GSD, or a foreign survey domain), exact localisation is neither achievable nor the objective. The only realistic task is *screening*: flag frames that plausibly contain an animal at high recall for a small false-alarm budget, so a human expert reviews a short queue instead of the whole flight. The right metric is recall at a fixed number of false positives per image (recall@FPPI , an FROC curve). The real anchor here is an out-of-domain aerial marine-mammal survey (Section 4), where the detector reaches useful but bounded triage recall.

In plain terms: two axes, five tests. Two things change independently in an aerial survey, and conflating them is the usual source of misleading accuracy numbers. The first is *scale* — how many pixels the animal actually spans, set by altitude and sensor rather than by the animal itself. The second is *domain* — whether a frame *looks like* the imagery the detector was trained on (same style of sensor, sea state and viewpoint) or comes from a different survey it has never seen. Every experiment in this report is deliberately one cell of the resulting 2×2 — scale (resolved vs. a few pixels) $\times$ domain (in-distribution vs. a foreign survey) — each chosen to isolate a single effect. Table 1 places all of them on both axes at once, so the reader does not have to reassemble the picture from scattered sections: it separates what tiling *can* fix (a *scale* problem) from what it cannot (a *domain* problem).

The same trained weights serve both ends; what changes along the axis is the *inference geometry* (Section 5) and the appropriate criterion by which success is judged. Every component below supports this claim: the curation pipeline controls what the detector learns (Section 3); a controlled six-way study isolates which data/capacity choices move the low-GSD screening

floor (Section 4); adaptive SAHI restores the high-GSD exact-detection regime a fixed model cannot reach (Section 5); and an explicit discussion of the domain gap (Section 6) bounds what the low-GSD end can promise today.

Contributions. This report makes the following contributions:

1. A reproducible **data-curation pipeline** (Section 3) that unifies ten heterogeneous public aerial and surface sources into a single deduplicated dataset, resolving label-format, EXIF-orientation and offline-augmentation pitfalls that silently corrupt naively merged training sets, plus a Label Studio human-in-the-loop pseudo-labelling loop for dataset growth.
2. A framing of the deployment problem as a **GSD axis with two tasks and two metrics** (exact detection via mAP; screening via recall@FPPI), and a **controlled six-way study** (Section 4) that isolates the effect of backbone capacity, surface data, the porpoise source and hard-negative dose on each — showing that composition and capacity, not tuning, govern the low-GSD screening floor.
3. **Adaptive SAHI** (Section 5), a telemetry-driven slicing scheme that couples tile size to GSD so the detector observes animals at a consistent apparent scale, with a fall-back cascade for missing metadata, validated on a semi-synthetic altitude sweep and on a real 151-megapixel survey frame.
4. An explicit **domain-gap analysis** (Section 6) that separates the pure GSD effect (isolated on synthetic data) from the domain-shift effect (a foreign real survey), and states the resulting recall ceiling as an explicit limitation.
5. A **ROS 2 deployment node** (Section 7) that runs the detector on live or replayed imagery under a selectable inference strategy (fixed, adaptive or adaptive-range), reusing the exact offline inference core and publishing standard `vision_msgs` detections into the AEROSUB autonomy stack.

Table 1: Experimental roadmap. Every evaluation in this report placed on both axes at once. *Scale* = how many pixels the animal spans (set by altitude and sensor); *Domain* = in-distribution vs. a foreign survey the detector was not trained on. Reading them together separates the *scale* effect (addressed by adaptive SAHI, Section 5) from the residual *domain* gap (addressed by in-domain pseudo-labelling, Section 6).

Test set	Scale	Domain	Task / metric	What it isolates
Aerial slice	resolved	in-dist.	exact / mAP	ceiling when nothing shifts (mAP_{50} 0.82–0.86)
whale-bbox-layer	resolved	foreign	exact / mAP	pure domain gap; SAHI cannot help (mAP_{50} 0.54)
Synthetic sweep	varied	in-dist.	recall / tiles	pure scale effect — adaptive SAHI’s job
HiDef survey	tiny	foreign	screening / recall@FPPI	scale + domain together ($\approx$ 0.6 ceiling)
APEM frame	tiny→resolved	foreign	exact / TP–FP	SAHI restores scale (3/3)

Scale is the solved axis (adaptive SAHI); domain is the residual ceiling (pseudo-labelling).

2 Related Work

Aerial marine-mammal detection. Automated detection of marine mammals in aerial imagery has been explored with classical and deep approaches [1–5]. Early systems used hand-crafted features and sliding windows; recent work uses CNN detectors on drone and manned-survey imagery. The recurring difficulties are extreme scale variation of targets (from a few

pixels to a large fraction of the frame), confusion with sun glint, whitecaps and wakes, and the scarcity of labelled data — all of which we observe and address here.

Digital aerial surveys and the GSD ceiling. The operational context for this work is the *digital aerial survey*, now the standard method for offshore-wind wildlife baselines in the UK and Ireland. Two commercial providers dominate this market: HiDef Aerial Surveying Ltd.¹ flies high-altitude digital video and stills tuned for wide-area *screening*, while APEM Ltd.² operates very-high-resolution still cameras (Phase One medium format) cited as a UK benchmark for survey design. The two providers sit at opposite ends of the ground-sample-distance (GSD) axis that organises this paper — HiDef anchoring the low-GSD screening regime and an APEM-class 151-megapixel frame anchoring the high-GSD exact-detection regime (Section 5). Bartlett et al. [6], whose authors include members of the present team, report that at the GSD typical of such surveys an individual animal spans only a handful of pixels, so that automated pipelines are well suited to detecting and counting animals but exact species-level identification from a single frame is inherently limited — precisely the two-task, two-metric split we adopt here.

One-stage detectors. The YOLO family [7, 8] provides real-time single-stage detection suitable for on-board deployment. We use YOLO11 [9]; unlike much prior work that fixes on the smallest backbone for edge speed, we treat backbone capacity as an experimental variable (Section 4) and find it matters specifically for the low-GSD screening regime.

Small-object and screening inference. Survey frames are tens to hundreds of megapixels, but detectors ingest a fixed, much smaller input; naive downscaling destroys small targets. Slicing-Aided Hyper Inference (SAHI) [10] runs the detector on overlapping tiles and merges results, trading compute for recall on small objects. We build on SAHI and extend it with telemetry-driven tile sizing (Section 5). For video, multi-object tracking [11] stabilises detections. Because the low-GSD regime is a detection/screening task with a small target set and a tolerance for coarse localisation, we score it with a free-response operating characteristic (FROC): recall against false positives per image, a standard screening metric, rather than mAP alone.

3 Data, Curation, Labelling

3.1 Sources

We aggregate ten datasets confirmed to contain only cetacean targets, spanning nadir and oblique aerial views and near-surface imagery (Table 2). Sources arrive in incompatible formats: Roboflow-YOLO exports, plain per-scene YOLO text files, VIA polygon JSON (the Dryad humpback/blue-whale set), and NOAA-style sighting CSVs. Species range from large whales that fill much of the frame to porpoises only a few pixels across, giving a demanding multi-scale detection problem (Fig. 1). All annotations are collapsed to a single class (**cetacean**).

Data availability and scarcity. Open aerial-*nadir* cetacean imagery is scarce. Most publicly available marine-mammal datasets are surface- or oblique-view (whale-watching or boat-based) or close-range documentary footage, rather than the high-altitude top-down survey geometry AEROSUB targets. The two large-whale nadir sets (**aerial-right-whale**, **whale-bbox-layer**) are small Roboflow projects; the humpback/blue-whale set is a Dryad archive; the porpoise frames come from a low-altitude UAV survey; and further aerial footage was frame-extracted from public survey imagery and pseudo-labelled (Section 3.4). Every dataset source is public and linked from Table 2 (Roboflow Universe, Dryad, Zenodo and NOAA archives), under their

¹<https://www.hidefsurveying.co.uk/>

²<https://www.apemltd.com/>



Figure 1: Representative frames from eight of the dataset sources (ground-truth boxes in green), spanning the scale range: from surface and near-nadir whales that fill much of the frame (**cetacean-detector**, **whaleshape**, **dryad**, **aerial-right-whale**) to nadir survey whales and porpoises only tens of pixels across (**whale-bbox-layer**, **gommapps**, **uav-porpoises**, **dolphin**). This three-order-of-magnitude apparent-size spread is the multi-scale challenge quantified in Table 2.

Table 2: Confirmed cetacean sources merged into the dataset (post-normalisation, with the reviewed **gommapps** survey set folded in). “Median box” is the median box $\sqrt{\text{area}}$ in pixels; the three-order-of-magnitude spread across sources is the multi-scale challenge. Nadir = top-down; Oblique = angled. The *Access* column links each source to its public release (Roboflow Universe, Dryad, Zenodo or NOAA). The two held-out survey *anchors* at the foot are proprietary vendor frames (available on request) and are *not* counted in the dataset total.

Source	Domain / view	Images	Boxes	Median box (px)	Access
whaleshape	aerial, nadir	461	623	1366	Dryad
cetacean-detector	aerial	686	774	1710	Roboflow
dryad (humpback/blue)	aerial, oblique	378	510	1574	Dryad, Dryad
gommapps (survey)	aerial, nadir	420	1874	329	NOAA
aerial-right-whale	aerial, nadir	500	510	246	Roboflow
whale-bbox-layer	aerial, nadir	514	747	113	Roboflow
cetacean-detection	aerial	1,448	1,765	71	Roboflow
dolphin2	surface	488	769	99	Roboflow
dolphin	surface	1,005	1,403	49	Roboflow
uav-porpoises	aerial, nadir	21,390	46,188	41	Zenodo
Total		27,290	55,163	45	
<i>Held-out survey anchors (deployment domain; not in dataset total):</i>					
HiDef survey (screening)	aerial, nadir	704	838	$\lesssim 15$	HiDef
APEM frame (exact)	aerial, nadir	1	3	≈ 120	APEM

respective licences — the **uav-porpoises** set is CC-BY-NC and the **gommapps** frames are US-Government public domain, pseudo-labelled by us. Only the two HiDef/APEM survey anchors are proprietary vendor frames, available on request. We redistribute the normalisation and evaluation code together with these source links rather than a repackaged copy of the raw dataset.

3.2 Normalisation pipeline

A single normalisation step converts every source into a common layout (image symlinks + normalised YOLO labels + a global manifest) without ever mutating the raw data (Fig. 2). It resolves three pitfalls that otherwise silently corrupt a merged training set:

- **Content deduplication.** MD5 hashing removes byte-identical images that appear across sources, which eliminated cross-split leakage (e.g. Dryad images present in both train and

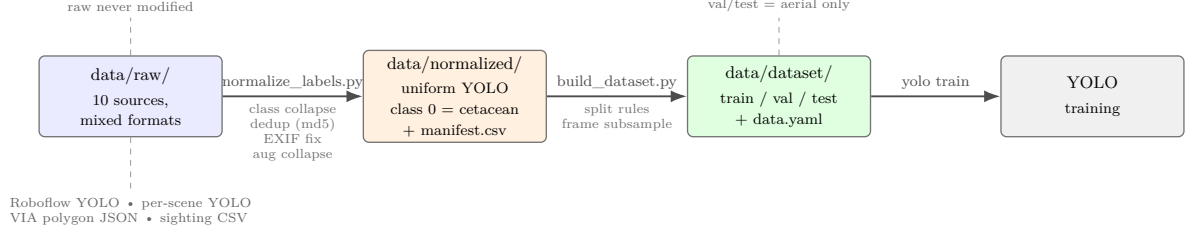


Figure 2: Data-preparation pipeline. Ten raw sources in mixed formats (Roboflow YOLO, per-scene YOLO, VIA polygon JSON, sighting CSV) are converted into a uniform single-class YOLO dataset (class collapse, MD5 deduplication, EXIF-orientation correction, offline-augmentation collapse), then a dataset builder materialises aerial-only validation/test splits. The raw data is never modified.

test) and repeated UAV frames.

- **EXIF-orientation correction.** Many Dryad images carry EXIF orientation flags 3/6/8 while their VIA polygon coordinates are in the *raw* pixel frame. The pipeline bakes the rotation into re-encoded JPEGs *and* applies the matching coordinate transform (rot180, rot90CW, rot90CCW), verified per-orientation. Every other source was confirmed orientation-1.
- **Offline-augmentation collapse.** Roboflow exports bake in offline augmentation (original + flip + brightness under one stem). These were collapsed to one instance per original image, which byte-level dedup misses since flips change the bytes. Only training splits are affected; val/test are left intact.

After normalisation the dataset contains 27,290 **images** / 55,163 **boxes** / 640 **background (negative) frames**. A dataset builder then materialises train/val/test splits with the invariant that *validation and test are aerial-only*, that the **whale-bbox-layer** source is held out *whole* as a precision test (no leakage of that domain into training), and that UAV video scenes are split-disjoint and subsampled at scene level to avoid near-duplicate frames leaking across splits.

3.3 Scale analysis: a bimodal, porpoise-weighted dataset

Analysing box sizes both sets the tiling strategy and previews the GSD-axis argument. The apparent-size distribution is strongly *bimodal* (Fig. 3): a porpoise / small-delphinid mode near 40 px, supplied overwhelmingly by **uav-porpoises** (84% of all boxes), and a whale / large-dolphin mode three orders of magnitude larger (medians of 1300–1700 px for **whaleshape**, **cetacean-detector**, **dryad**). The global median box $\sqrt{\text{area}}$ is ≈ 45 px, landing in the sparse valley *between* the two modes — a subtlety that matters when tuning inference (Section 5), since a single magnification target set at the global median serves *neither* mode well. Because most sources contain large in-frame animals and fewer than a per cent of boxes fall below ~ 10 px at 1024 px input, tiling is deferred to the *inference* stage rather than baked into training; the working SAHI tile is set by the small end (porpoise/screening regime) at 640 px, overlap 0.2, model input 1024 px.

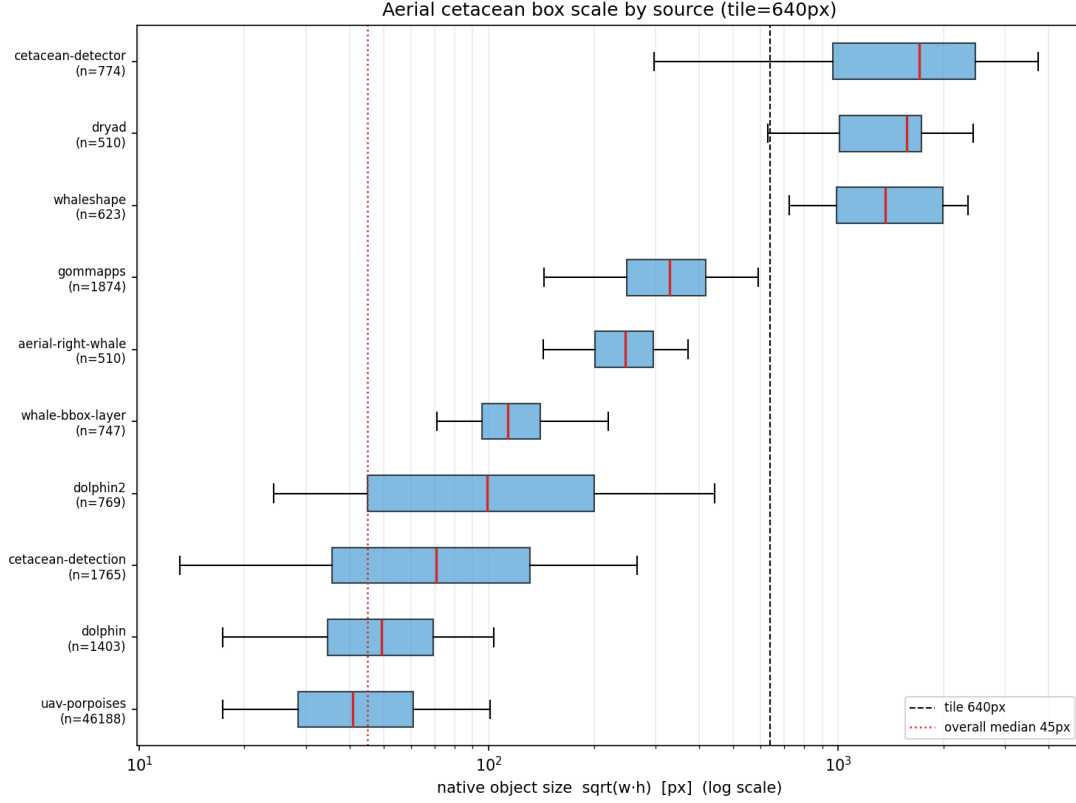


Figure 3: Box-size distribution across sources (box $\sqrt{\text{area}}$, log scale). The dataset is strongly *bimodal*: a dominant porpoise / small-delphinid mode near 40 px (**uav-porpoises**, 84% of boxes) and a whale / large-dolphin mode three orders of magnitude larger. The global median (≈ 45 px) sits in the valley between the modes. This three-order-of-magnitude spread within one single-class dataset is the physical basis of the GSD axis: no single input scale can serve every altitude, which motivates resolving scale at inference time (Section 5).

A porpoise-weighted dataset is representative, not biased. This porpoise-heavy composition is not a sampling artefact: harbour porpoise and small delphinids numerically dominate real North-East Atlantic shelf populations, whereas large baleen whales are comparatively rare [12]. A dataset — and a benchmark — weighted toward small targets is thus *representative* of the operational species mix, and the small-target recall floor it exposes is precisely the one that matters most in deployment. Section 4 shows that this source is also critical: removing it collapses the detector.

3.4 Label Studio pseudo-labelling loop

Labelled aerial cetacean data is scarce, but unlabelled aerial footage is easy to obtain. We close this gap with a human-in-the-loop pseudo-labelling loop (Fig. 4) built on Label Studio [13]:

1. **Predict.** The current best model is run over an unlabelled aerial pool (at low confidence so the human mostly corrects rather than draws) to produce pre-annotations.
2. **Review.** A human accepts, corrects or rejects each box, keeping *cetaceans only* — an essential filter, because the source sighting logs also contain turtles, sharks and even “whale shark” (a fish) — and applying a floor rule that skips ambiguous, faint low-contrast targets.
3. **Merge.** Reviewed labels are folded back as a new normalised source and appended to the manifest.
4. **Retrain.** The detector is retrained on the *combined* set (never new-only, which causes forgetting), and a fixed held-out test set enables round-over-round comparison.

The reviewed **gommapps** survey set (420 reviewed frames / 1,874 boxes / 57 empty) enters the dataset this way and contributes both positives and, from its no-sighting frames, hard negatives (Section 4). The loop provides the scalable mechanism by which future AEROSUB survey flights become training data.

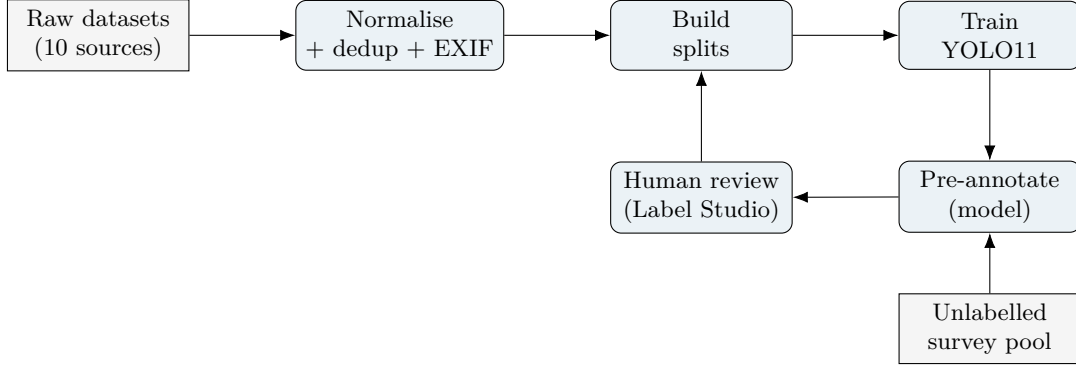


Figure 4: Data and pseudo-labelling pipeline. Curated datasets train the detector; the detector pre-annotates unlabelled survey footage; a human reviewer corrects the labels; corrected data re-enters the combined training dataset.

4 The Detector and the Screening Study

4.1 Detector and training

We use **YOLO11** [8, 9], a single-class detector at input resolution 1024 px. Training uses the Ultralytics framework (v8.4) on a single Quadro RTX 4000 (8 GB): batch 18, up to 100 epochs with early stopping (patience 20), fixed seed 1337. We keep augmentation view-agnostic — vertical flips and large rotations are label-preserving only for nadir imagery and corrupt oblique frames, so the global schedule uses horizontal flip, HSV, scale, translate and mosaic only. A view-conditional strategy is left as future work. The baseline M1 converges cleanly under this schedule: training and validation losses fall in step (no over-fitting) and validation mAP plateaus by epoch ~ 80 (Fig. 5).

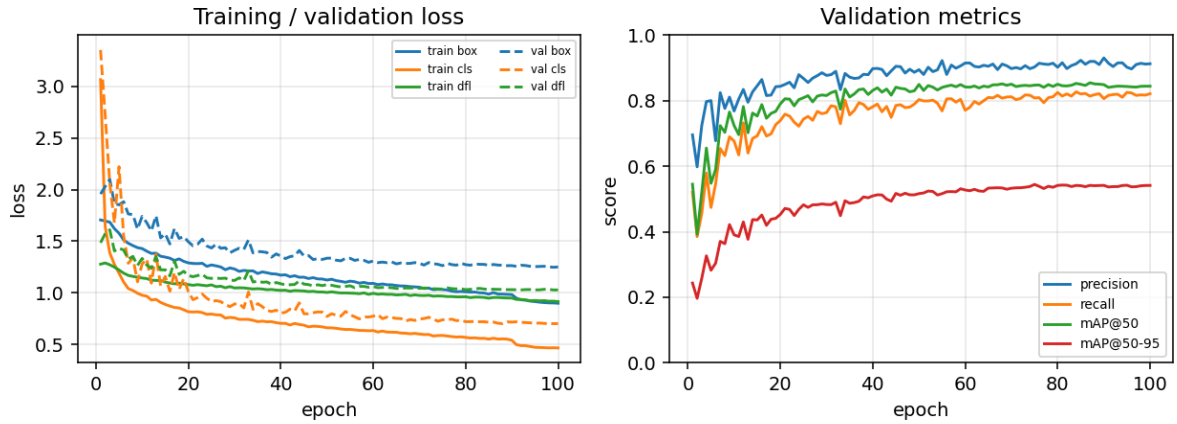


Figure 5: Baseline detector (M1, YOLO11n) training. Left: training (solid) and validation (dashed) box/cls/dfl losses fall together, indicating no over-fitting. Right: validation precision, recall and mAP rise and plateau ($mAP50 \approx 0.84$, $mAP50-95 \approx 0.54$ at epoch 100).

Two operating points, two metrics. Following the GSD-axis argument, we evaluate the detector at both ends of the axis on fixed, held-out data:

Table 3: The six controlled training configurations. Each changes one factor from baseline **M1**. Dose = negatives as a fraction of the final training set; *Train img.* is the number of images in that config’s training split. All evaluated on the identical fixed test sets. Note how removing the porpoise source (M4) shrinks the training set threefold — a first sign of how indispensable it is.

Variant	Backbone	Train img.	Change from M1
M1	YOLO11n	5,394	baseline (aerial, porpoises in, 10% neg)
M2	YOLO11s	5,394	<i>capacity</i> $\uparrow$ (same data as M1)
M3	YOLO11n	8,618	+ four surface sources
M4	YOLO11n	1,506	porpoise source <i>removed</i>
M5	YOLO11n	4,855	hard-negative dose 0%
M6	YOLO11n	6,069	hard-negative dose 20%

- **High-GSD, exact detection.** A held-out *source* (**whale-bbox-layer**, 106 images / 153 instances, never seen in training) measures precision and mAP at $\text{IoU} \geq 0.5$ under plain inference. An in-distribution aerial slice (291 images, porpoise-dominated) is reported alongside.
- **Low-GSD, screening.** An external HiDef Aerial Surveying digital aerial survey (denoted the *HiDef* test set; 704 frames / 838 boxes / 12 empty, fully held out, a *different* survey domain; Fig. 6) measures recall against false positives per image (recall@FPPI, $\text{IoU} \geq 0.3$ one-to-one match) under SAHI tiling — the screening/FROC metric.

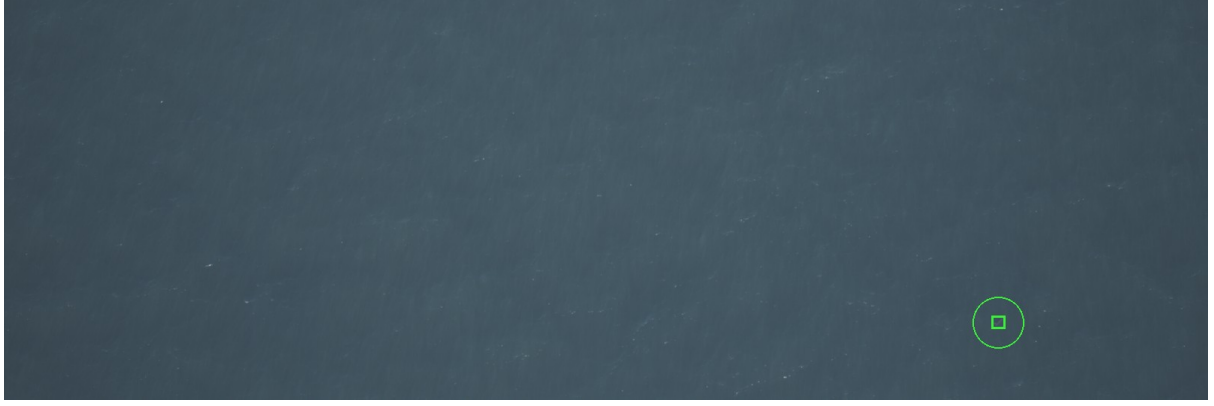
A controlled six-way study. To find which choices move each operating point we change *one* factor at a time from a baseline **M1** (YOLO11n; aerial sources; porpoises in; 10% hard-negative dose), holding seed and schedule fixed (Table 3).

4.2 High-GSD result: exact detection

Table 4 reports the precision regime. On the held-out **whale-bbox-layer** source the models cluster tightly ($mAP_{50} \approx 0.50\text{--}0.54$), and *capacity barely moves it*: the YOLO11s variant M2 (mAP_{50} 0.540) is within a point of the YOLO11n baseline M1 (0.538). Removing porpoises (M4) is the one large effect, and it is strongly negative (mAP_{50} 0.202): even for large-whale precision, the porpoise source is critical. Extra surface data (M3) slightly hurts, and hard-negative dose trades a little precision for a little recall. The takeaway of the high-GSD end is that, once the animal is resolved, this task is capacity-insensitive and configuration-robust — backbone capacity is not the bottleneck.

Table 4: High-GSD **exact-detection** regime, plain inference at 1024 px. Held-out source = **whale-bbox-layer** (106 img / 153 inst); aerial slice = in-distribution (291 img, porpoise-dominated). Capacity (M2) barely changes mAP; removing porpoises (M4) collapses even large-animal precision.

Variant	Backbone	Held-out source		Aerial slice	
		mAP50	mAP50-95	mAP50	mAP50-95
M1	11n	0.538	0.221	0.824	0.520
M2	11s	0.540	0.227	0.863	0.550
M3	11n	0.510	0.238	0.793	0.508
M4	11n	0.202	0.070	0.143	0.093
M5	11n	0.531	0.212	0.823	0.532
M6	11n	0.503	0.224	0.801	0.521



(a) A HiDef survey frame (downscaled to fit the column): one confirmed harbour porpoise (green box, ringed for visibility) in an otherwise empty sea.



(b) Magnified crop of that porpoise: only ≈ 50 px across at the survey’s native resolution — a faint, low-contrast target.

Figure 6: The low-GSD *screening* regime on the held-out HiDef Aerial Surveying digital aerial survey. At this ground-sample distance a harbour porpoise occupies only a few tens of pixels in a multi-megapixel frame, so exact species identification from a single frame is impossible and the appropriate task is high-recall triage: cueing a human expert to the frame (and tile) that contains an animal.

4.3 Low-GSD result: wide-area screening

The screening regime shows the opposite pattern (Table 5, Fig. 7). Here *capacity matters*: the YOLO11s variant M2 lifts recall at 1 false positive per image from M1’s 0.412 to 0.535 — a 30% relative gain — and dominates at the tight operating points (0.240 vs 0.063 recall at 0.1 FPPI). The larger backbone helps exactly where the target is a few low-contrast pixels and shape must be inferred, which is the opposite of the common “nano is enough for edge detection” assumption. Composition again dominates the rest:

- **Porpoises are essential.** M4 (porpoises removed) is flat on the floor (≤ 0.04 recall at every budget): a detector that never learned that a faint, low-contrast target can be an animal cannot screen at low GSD.
- **Surface data hurts.** M3 (+surface) sits well below M1/M2; boat-deck and close-range imagery pulls the detector away from the nadir screening regime.

Table 5: Low-GSD **screening** regime on the out-of-domain HiDef survey (704 frames), SAHI tile 1024 px. Box-level recall at a false-positive budget ($\text{IoU} \geq 0.3$). Best per column in bold. Capacity (M2) and porpoise inclusion govern the floor; M4 collapses.

Variant	R@0.1	R@0.5	R@1.0	R@2.0
M1	0.063	0.264	0.412	0.573
M2	0.240	0.449	0.535	0.611
M3	0.132	0.260	0.307	0.384
M4	0.004	0.016	0.024	0.038
M5	0.081	0.396	0.525	0.635
M6	0.033	0.173	0.296	0.443

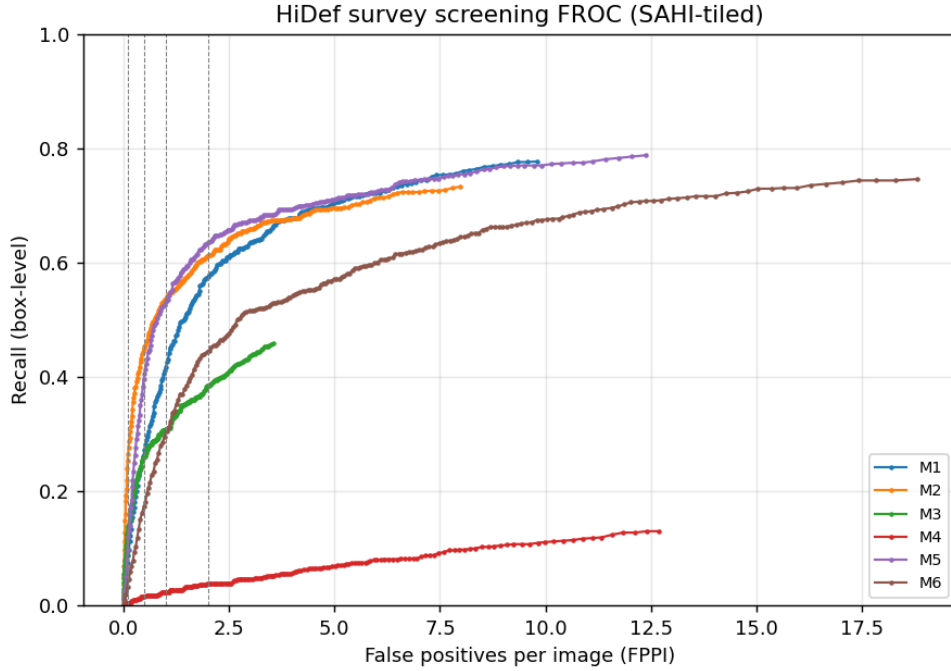


Figure 7: Screening FROC on the out-of-domain HiDef survey (SAHI tile 1024 px): box-level recall vs. false positives per image. The larger-capacity M2 (orange) and the negative-free M5 (purple) lead; the porpoise-free M4 (red) never leaves the floor. Dashed verticals mark the 0.1/0.5/1/2 FPPI budgets of Table 5. Recall plateaus near 0.6–0.65 — the domain-gap ceiling of Section 6.

- **Hard negatives trade recall for precision.** In the recall-oriented screening regime *fewer* negatives is better at loose budgets: M5 (0%) reaches 0.635 recall at 2 FPPI versus M1’s 0.573 and M6’s 0.443. Hard negatives buy precision at a cost in the recall we care about here.

We also find that the coarser SAHI tile (1024 px) is both more accurate for the strong configurations *and* $\approx 28\%$ cheaper than 640 px on these frames, so it is the preferred screening operating point.

What the detector catches — and misses. Figure 8 makes the screening operating point concrete. At 2 FPPI the best-performing M2 recovers 513/838 animals (0.61 recall): despite being trained overwhelmingly on much larger whales, it reliably fires on harbour porpoises only ≈ 50 apparent pixels across in a foreign survey domain. Crucially the missed animals are *not* smaller than the detected ones — both have a median box of ≈ 50 px (detected 49.4 px, missed 50.2 px) — so the low-GSD limit is set by *contrast*, not raw size: the misses are the faintest, lowest-contrast targets, animals under glare or against dark, wake-broken water. Recovering

three in five of these targets is a strong screening result, and it is why more in-domain data (Section 3.4), rather than a larger input, is the lever that raises the floor.

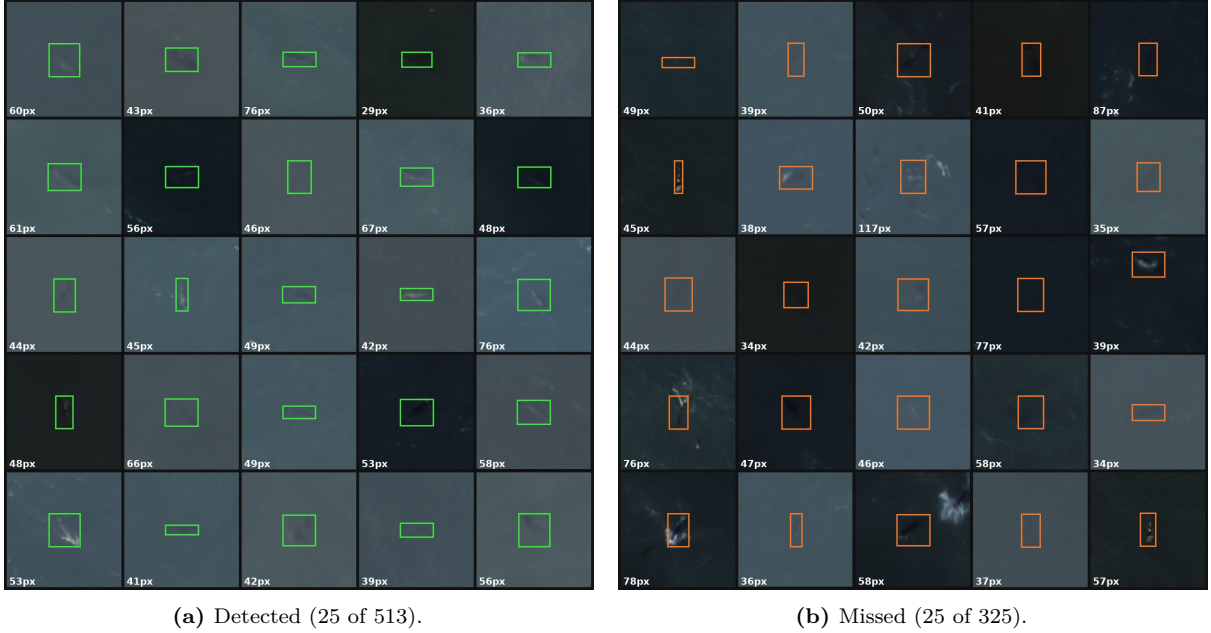


Figure 8: Best-performing M2 on the HiDef survey at the 2 FPPI operating point (0.61 recall); each crop is centred on a ground-truth porpoise and labelled with its native box size in pixels. (a) Detected and (b) missed animals are the *same* apparent size (median ≈ 50 px); the misses are simply the lowest-contrast cases — faint, low-contrast targets under glare or against dark, wake-broken water — not smaller animals.

Summary. The two operating points of one detector behave oppositely under the same changes: high-GSD exact detection is capacity-insensitive and configuration-robust, while low-GSD screening is capacity- and composition-sensitive. The best-performing configuration overall is **M2** (YOLO11s), which leads screening outright and matches the field at exact detection; it is used for the scale-adaptive inference experiments below.

5 Scale-Adaptive Inference

Sections 3.3 and 4 establish that the target’s *apparent* size is the dominant variable. This section closes the loop with inference geometry that holds apparent size roughly constant across the GSD axis.

5.1 Why a fixed slice is rarely optimal

Plain inference downscales the whole frame to 1024 px, so a target’s apparent size depends on field of view (altitude, lens, sensor), not on the sensor’s megapixel count. Extra megapixels help *only through* tiling: on a 151-megapixel survey frame, plain inference detects nothing (Section 5.5); the animals are downscaled below the detection floor. SAHI [10] runs the detector on overlapping tiles and merges the results, but a *fixed* tile size is rarely optimal: if tiles are too large the animal is downscaled away; if too small, compute explodes and whitecaps and glint are over-magnified into a large number of false positives. Two practical SAHI findings recur throughout: the default confidence of 0.25 is unreliable on water and must be raised toward 0.5, and SAHI’s default non-maximum merging (NMM) reconciles overlapping detections across tile seams more reliably than plain NMS.

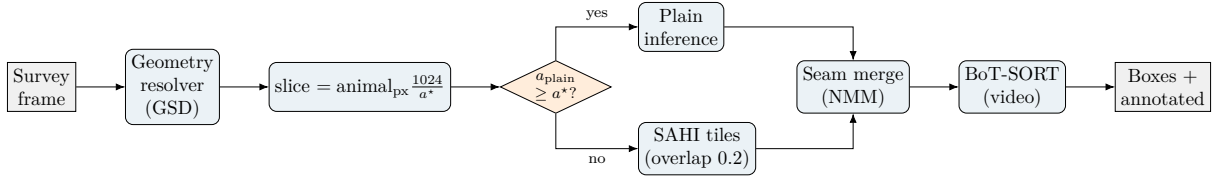


Figure 9: Adaptive SAHI inference pipeline. Flight geometry sets the crop size; frames whose animals are already detectable at plain resolution skip tiling entirely, and video streams add BoT-SORT tracking after box merging.

5.2 Adaptive SAHI: the scale solver

Adaptive SAHI uses flight geometry to size tiles so a target is re-magnified to a roughly constant apparent size at the model input, and *skips* tiling entirely when the animal is already large enough. Given camera pixel pitch p , focal length f and above-water altitude h , the ground-sample distance and native target size are

$$\text{GSD} = \frac{p h}{f}, \quad \text{animal}_{\text{px}} = \frac{L_{\text{target}}}{\text{GSD}}, \quad (1)$$

where L_{target} is the expected animal length. A survey frame of long side W is downsampled by $W/1024$ before plain inference, so the detector sees the animal at $a_{\text{plain}} = \text{animal}_{\text{px}} \cdot 1024/W$ pixels. The solver uses one apparent-size target a^* (the trained detection scale; here $a^* \approx 45$ px for a nominal $L_{\text{target}} = 3$ m):

- *Skip rule.* If $a_{\text{plain}} \geq a^*$ the animal is already detectable, so SAHI is skipped for a single plain pass.
- *Slice rule.* Otherwise the crop size that renders the animal at a^* at model input 1024 is $\text{slice} = \text{animal}_{\text{px}} \cdot 1024/a^*$.

For the 151-megapixel frame below (GSD 2.41 cm/px, so $\text{animal}_{\text{px}} \approx 125$ px for a 3 m animal) this yields $\text{slice} \approx 2834$ px — large, few tiles, yet still magnifying the pod far above the plain-inference floor.

A single length or a size range. Given one length the solver sizes a single slice for that scale. Given a size *range* $[L_{\min}, L_{\max}]$ it instead sizes the crop so the *largest* expected animal still fits within one tile, $\text{slice} = L_{\max}/(\text{GSD} \cdot \phi)$ with fill fraction $\phi \leq 1$. This *range-aware* variant stops large whales being fragmented across tile seams at low altitude, at the cost of more tiles; the single-length policy is cheaper when the population is uniform.

5.3 Scale-anchor cascade

The solver needs one scale anchor, resolved by a fall-back cascade (Fig. 10) ordered from most to least reliable so the system is autonomous whenever possible: (1) an explicit operator override; (2) a user-/survey-supplied GSD or drone telemetry (DJI XMP `RelativeAltitude`, or EXIF altitude); (3) a per-flight camera model (pixel pitch, focal length) plus altitude, giving GSD from the equation above; (4) with no geometry, a blind sweep of 512/1024/2048 keeping the most self-consistent result. All four tiers are implemented and used below; a fifth self-calibration tier — probing a sparse subset and back-solving GSD from the first confident detection — is left as future work.

5.4 Controlled altitude sweep

Evaluating altitude behaviour needs matched frames of the *same* animals at many altitudes with exact ground truth — data that does not exist for wild cetaceans. We therefore build a *semi-synthetic altitude sweep*: 18 real, model-verified animal cut-outs (lengths 1.7–18.2 m, a realistic

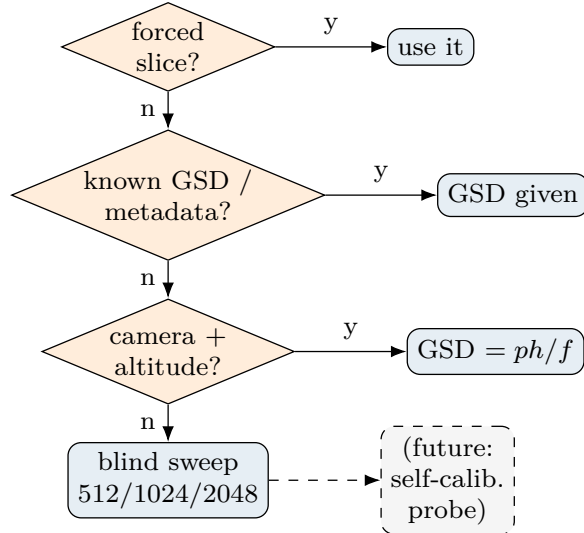


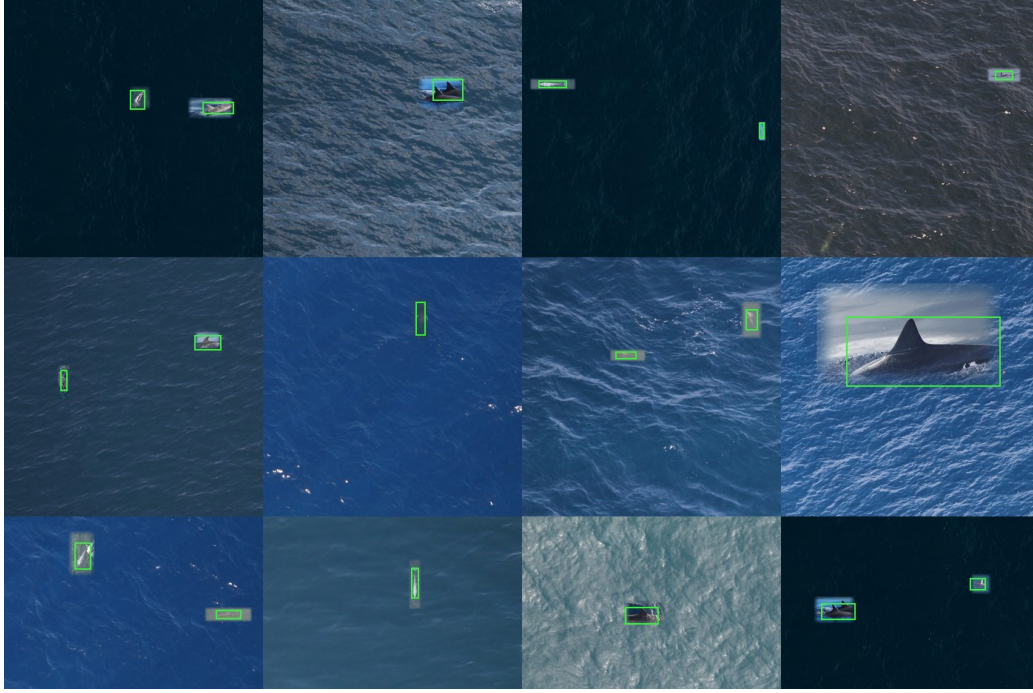
Figure 10: Scale-anchor cascade. Each answered branch resolves GSD (hence the crop size); the self-calibration probe (dashed) is left for future work.

Table 6: Altitude sweep, best-performing M2: recall by altitude and tiles/frame per method. Adaptive spends tiles only as geometry demands and *approaches* fixed-slice recall at a fraction of the cost, though fixed-640 keeps a recall edge; *range-aware* closes most of the gap at high altitude. At 60 m the adaptive skip rule (reasoning about a nominal 3 m animal) reverts to plain and loses the smaller animals — a real, tunable limit (see text).

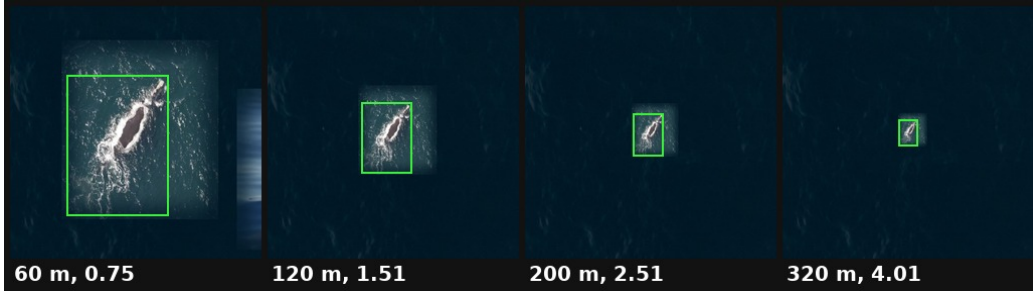
Alt. (GSD)	Plain (1)	Fix-640 (176)	Fix-1024 (70)	Adaptive (1–24)	Range-ad. (1–280)
60 m (0.75)	0.18	0.88	0.71	0.18	0.18
120 m (1.51)	0.11	0.50	0.50	0.22	0.39
200 m (2.51)	0.17	0.39	0.44	0.22	0.44
320 m (4.01)	0.06	0.22	0.17	0.11	0.11

porpoise-weighted mix) are feathered onto a real hard-negative ocean canvas (8192×5460 , stitched whitecap/glint tiles) and geometrically resampled — downscale only — to the exact size each animal subtends at seven altitudes (60–320 m) under DJI Zenmuse P1 geometry ($p = 4.39 \mu\text{m}$, $f = 35 \text{ mm}$), so the *same* animals shrink with altitude on an unchanged sea texture (Fig. 11). This isolates the pure GSD variable while keeping textures photorealistic. For each frame we compare plain inference, fixed-640/1024 SAHI, the single-length adaptive policy and the range-aware variant, scoring recall, tile count and latency at $\text{IoU} \geq 0.3$. Adaptive SAHI is an inference-time wrapper independent of the trained weights: we drive it with the best-performing M2 here, and additionally run baseline M1 on the real APEM frame (Table 7) to confirm the geometry-matched behaviour is not specific to one model (Table 6, Fig. 12).

A real limit at low altitude. At 60 m the adaptive skip rule fires: a nominal 3 m animal already subtends ≈ 50 apparent pixels after the plain downscale, so the policy reverts to a single plain pass and loses the sub-3 m animals that fixed-640 still recovers (0.18 vs 0.88). This is not a fundamental failure but a tuning choice: setting L_{target} (or the skip threshold) to the *smallest* expected animal restores fine tiling at low altitude, at the cost of more tiles. The default is deliberately conservative about compute; the point of the sweep is that adaptive *approaches* fixed recall at mid/high altitude for far fewer tiles — its advantage here is compute, not accuracy — and that the residual high-altitude decay is a scale-recovery limit shared by every method, not a scheduling one.



(a) One synthetic frame (120 m): 18 model-verified cut-outs (green) feathered onto a real whitecap/glint canvas.



(b) The *same* animal at four altitudes (label: altitude, GSD in cm/px): only the apparent size changes.

Figure 11: Semi-synthetic altitude sweep. Real animal cut-outs are composited onto a real hard-negative ocean canvas and downsampled to the size each subtends at a given altitude, isolating ground-sample distance from every other variable.

5.5 Real high-GSD anchor: a 151-megapixel survey frame

We anchor the high-GSD, exact-detection end of the axis on a real survey frame the detector has never seen: a 151-megapixel APEM digital-aerial-survey frame (Phase One iXM-RS150F, 14204×10652 px) of three bottlenose dolphins, with a recorded EXIF altitude of 448.5 m giving $\text{GSD} = 2.41 \text{ cm/px}$ — real geometry, not an assumption. We report a single APEM frame because the provider’s survey imagery is proprietary and we were unable to obtain their dataset; this publicly available frame is included as a representative high-GSD sample and to show the wide variation in GSD and image quality *between* survey vendors — contrast the coarse, low-contrast HiDef screening frames of Fig. 6 with this APEM stills frame, in which each dolphin is cleanly resolved. Running the best-performing M2 at the slice its geometry anchor dictates (Table 7, Fig. 13):

- **Plain @1024: 0/3.** After the $\sim 14\times$ whole-frame downscale each dolphin is a handful of pixels; the detector finds nothing, even though the animals are clearly resolvable in the raw frame. Megapixels help only through tiling.
- **Fixed-640: 2/3 with 8 false positives, 588 tiles.** Fine tiles over-magnify whitecaps into porpoise-shaped false targets (the baseline M1 produces 83 false positives here), and

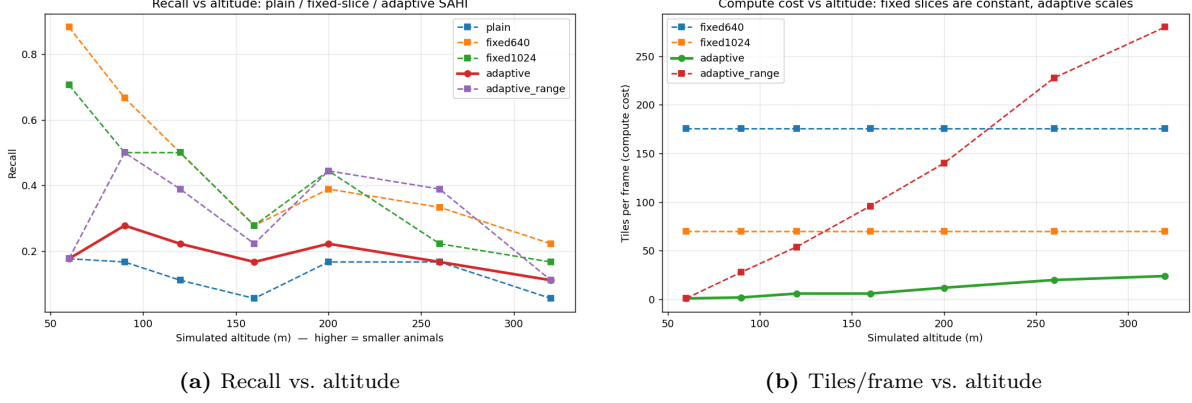


Figure 12: Altitude sweep (best-performing M2). Adaptive approaches fixed-slice recall at far fewer tiles (left; it does not exceed fixed slicing on recall) while spending tiles only as altitude demands (right), where fixed slicing pays a flat cost throughout. Above ~ 300 m recall falls for *all* methods: once native detail is lost, no crop can restore it.

Table 7: Real 151-megapixel APEM frame (three bottlenose dolphins, GSD 2.41 cm/px from EXIF). Recall (TP/FP) and tiles per method, $\text{IoU} \geq 0.3$. Plain sees nothing; adaptive is complete, clean and cheapest. Baseline M1 shown for the whitecap-false-positive contrast at fine tiles.

Method	Tiles	M2: R (TP/FP)	M1: R (TP/FP)
Plain	1	0.00 (0/0)	0.00 (0/0)
Fixed-640	588	0.67 (2/8)	0.67 (2/83)
Fixed-1024	234	0.67 (2/0)	1.00 (3/23)
Adaptive	35	1.00 (3/0)	1.00 (3/1)

the very finest pass still misses the faintest dolphin.

- **Adaptive (slice 2834): 3/3 with 0 false positives, 35 tiles, ≈ 9 s.** Sizing the crop to the EXIF GSD recovers all three dolphins with no whitecap false alarms, at $17\times$ fewer tiles than fixed-640 and the fastest wall-clock of any method.

This is the key result of the high-GSD end: geometry-matched tiling turns an apparently empty 151-megapixel frame into a clean, complete detection, and does so more cheaply than brute-force fine tiling because it magnifies the water just enough to expose the animals and no further.

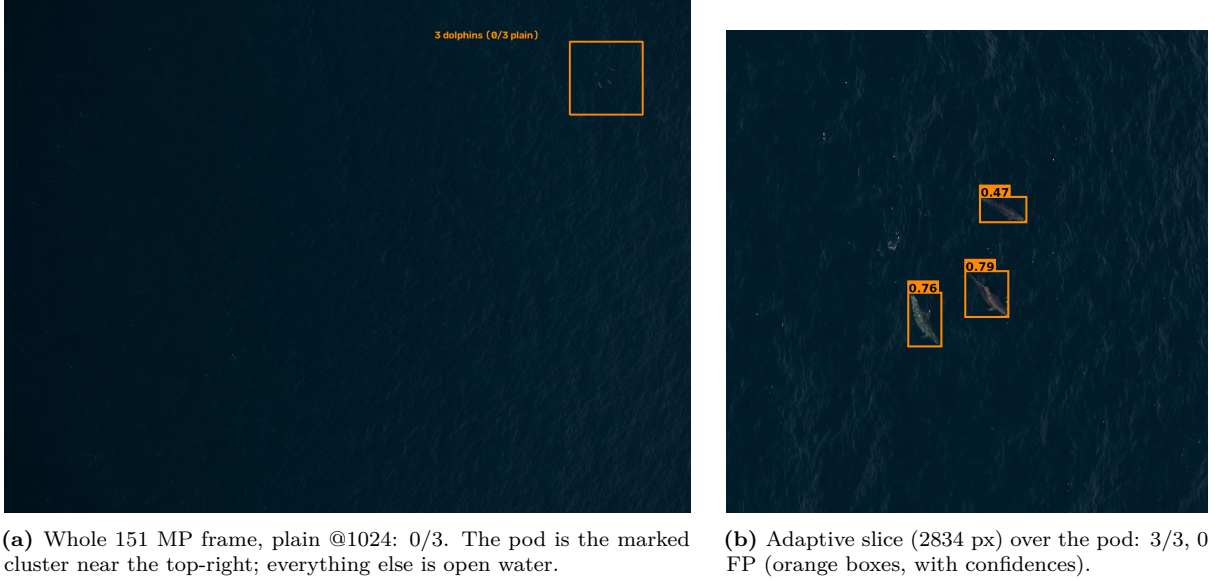


Figure 13: Real 151-megapixel APEM survey frame (Phase One iXM-RS150F, GSD 2.41 cm/px). Whole-frame plain inference (left) detects none of the three dolphins after the $\sim 14\times$ downscale, even though the pod (marked) is clearly resolvable in the raw pixels; adaptive SAHI, sized to the EXIF geometry, recovers all three in a single geometry-matched crop (right) with zero whitecap false positives, at 35 tiles.

6 The GSD Axis and the Domain Gap

The two real anchors bound the same detector at opposite ends of the GSD axis: adaptive inference recovers the full three-dolphin pod on the single high-GSD APEM frame, while at low GSD on the out-of-domain HiDef survey the best configuration plateaus near 0.6 recall at 2 FPPI (Fig. 7). Both are direct measurements of one model; the gap between them is the subject of this section.

Separating GSD from domain shift. The low-GSD screening ceiling confounds two effects: the animals are small *and* the survey is a different domain (different sensor, look and sea state) from the training sources. Our two other experiments deliberately separate them. The semi-synthetic sweep (Section 5.4) varies *only* GSD on in-domain textures and shows that recall degradation with altitude is a scale-recovery effect shared by all inference methods — pure GSD. The HiDef survey adds domain shift on top of low GSD, and the extra drop (recall capped near 0.6 even with tiling) is the *domain-gap* component. In other words, we train on one set of aerial domains and deploy on a different one, and the HiDef survey is a stress test of that transfer, not a like-for-like test.

Why this is the right way to report it. Reporting a single aggregate number would hide this structure. Framed along the GSD axis, the results are actionable: at high GSD the system is deployment-ready and its remaining work is engineering (on-board packaging, Section 7); at low GSD it is a *screening aid* with a stated recall of ≈ 0.5 – 0.6 at a low false-alarm budget — enough to cut an expert’s review queue by roughly half, but not a replacement for the observer. The levers that raise the low-GSD floor are now known and quantified: more capacity (M2), the porpoise source (M4), a low hard-negative dose (M5), and, above all, more in-domain labelled data via the pseudo-labelling loop (Section 3.4).

Limitations. Three caveats bound these claims. (i) The high-GSD accuracy result is a *single-frame demonstration*: the 3/3, 0-FP outcome rests on one real APEM frame (three dolphins) plus a semi-synthetic sweep, because provider survey imagery is proprietary (Section 5.5);

broader validation across many real high-GSD frames and surveys remains future work. (ii) On the synthetic sweep, adaptive SAHI’s demonstrated benefit is tile efficiency at *comparable* recall, not higher recall than fixed slicing (Table 6); the accuracy advantage is shown only on the single APEM frame. (iii) Every configuration is a single training run at one seed (1337) with no confidence intervals, so single-run deltas (e.g. the $\approx 30\%$ M2 screening gain) are indicative, not statistically established.

7 Deployment: ROS 2 Integration

A perception model is only useful once it runs on the aircraft. Because the AEROSUB UAS payload and ground segment are built on the Robot Operating System (ROS 2), the detector and the adaptive-SAHI pipeline of Section 5 are packaged as a self-contained ROS 2 node (`cetacean_detector`, an `ament_python` package for ROS 2 Humble) that drops into the existing autonomy stack.

Shared core. The node does not re-implement inference. The geometry solver, scale cascade and SAHI runner are a single importable package (`cetacean.inference`) used unchanged by both the offline evaluation of Section 5 and the node, so the behaviour measured on the bench is the behaviour that flies — there is no second code path to drift out of sync.

7.1 Node design

A single node covers the whole GSD axis, and the *inference strategy* is chosen with one `inference_mode` parameter, leaving the operating point to the mission rather than the code:

- **fixed** — fixed-size SAHI tiling (a constant crop), for a known, stable altitude or a quick baseline;
- **adaptive** — the crop is sized per frame from geometry so a target body length lands at the trained object scale, skipping SAHI when the animal is already resolved (Section 5);
- **adaptive-range** — the crop is sized so the *largest* expected animal still fits one tile, keeping a whole size range detectable at once (Section 5.4).

The node **subscribes** to `sensor_msgs/Image` — a live camera, or a folder of stills, a video or a single survey frame replayed by a companion publisher node — and, optionally, to `std_msgs/Float32` relative-altitude and/or ground-sample-distance telemetry. The latest telemetry value is cached and drives the per-frame slice decision through the same cascade (Section 5), falling back to operator-set geometry or a one-off blind sweep when none is present. It **publishes** `vision_msgs/Detection2DArray` (boxes and confidences), a `std_msgs/Int32` count, and an optional annotated `sensor_msgs/Image` for live monitoring or ground-station review. Using standard `sensor_msgs/vision_msgs` messages — not a custom type — means any downstream ROS 2 component (trackers, geolocators, mission planners, recorders, RViz) consumes the output unchanged. A few-line NumPy image converter replaces `cv_bridge`, so the node builds against the same CUDA/PyTorch environment used for training.

7.2 Verification and deployment

Run through ROS, the node reproduces the offline behaviour exactly. Fed the 151-megapixel APEM frame with its EXIF geometry, **adaptive** mode resolves the same crop the bench solver chose (≈ 2834 px) and recovers all three dolphins; on low-altitude frames the same mode resolves to plain inference and skips tiling — the two anchors of Section 5 reproduced without changing a line. Deployment is deliberately simple: build once with `colcon`, point the model parameter at a trained checkpoint (or an exported ONNX/TensorRT engine), select a mode, and launch;

swapping models or retuning thresholds is a parameter change, not a code change. The node runs unchanged on a laptop for bench testing and on an on-board NVIDIA Jetson (Orin class) for flight, with the model exported to TensorRT for edge throughput — plain inference is real-time, while SAHI ranges from near-real-time to offline depending on the tile budget the chosen mode implies. Because detections are published on `vision_msgs`, geolocation from telemetry and closed-loop UAS re-tasking attach as separate nodes, matching the AEROSUB goal of fused detection-and-tracking that steers the aircraft toward confirmed sightings.

8 Discussion and Future Work

The analysis points to two concrete directions.

Closing the domain gap with pseudo-labelling. The single largest lever on the low-GSD ceiling is in-domain data. The Label Studio loop (Section 3.4) is the mechanism: several larger rounds focused on *hard*, diverse survey frames (small animals, glint, wake) should raise screening recall directly, and every confidently screened frame from deployment feeds straight back in. Concretely, the ≈ 0.6 recall ceiling reported here is a property of the *current* training data, not a fixed limit: as AEROSUB flights collect new survey imagery, labelling it through the loop and retraining the same detector on this in-domain data is the expected route to a higher ceiling.

Other near-term work. (1) A view-conditional augmentation schedule for nadir vs. oblique imagery; (2) the self-calibration cascade tier (probe a sparse subset, back-solve GSD); and (3) geometry-based size estimation (inverting GSD to turn a detection’s pixel length into a physical length on nadir frames). On-board packaging of the shared inference code is already realised by the ROS 2 node of Section 7; the remaining deployment work is hardware-target optimisation (NVIDIA Jetson Orin, TensorRT export) and integration on the airframe.

9 Conclusions

We presented the RGB perception component of SurveyLabs’ autonomous aerial cetacean-monitoring capability for AEROSUB, developed under a real constraint: open, top-down aerial cetacean imagery is scarce and fragmented. Our starting contribution is therefore a practical one — consolidating ten heterogeneous public sources into a single, deduplicated, correctly-oriented dataset of 27,290 images, and a human-in-the-loop pseudo-labelling loop to keep growing it. On this dataset we trained a single robust detector and analysed *where* it can be trusted, organising the study around the ground-sample-distance axis whose two ends are different tasks: exact detection at high GSD and wide-area screening at low GSD.

The analysis shows the approach works well with the data available. A controlled six-way study identifies the levers that actually move performance — porpoise data is critical, surface data hurts, fewer hard negatives help recall, and, reversing the usual assumption, the larger YOLO11s backbone lifts screening recall by 30% at negligible cost to exact-detection mAP. Scale-adaptive SAHI restores the high-GSD end, recovering all three dolphins in a real 151-megapixel frame with zero false positives at $17\times$ fewer tiles than fixed slicing, where plain inference sees nothing. Most importantly for deployment, the detector transfers to imagery it was never trained on: on an unseen survey domain — a different sensor, sea state and provider — it still screens roughly three in five animals at a low false-alarm budget, cutting the expert’s review queue by roughly half.

We state the remaining limit plainly: out-of-domain recall plateaus near 0.6 at 2 false positives per image, a domain gap rather than a scale problem, and the pseudo-labelling loop is

the direct route to close it as AEROSUB survey flights supply in-domain data. Taken together — a curated dataset from scarce sources, a characterised detector, scale-aware inference and a ROS 2 deployment node — these artefacts form a working foundation for on-board cetacean monitoring, and the basis of AEROSUB deliverable D2.1 and open-access key exploitable result KER9.

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References

- [1] Frederic Maire, Luis Mejias, and Amanda Hodgson. Automating marine mammal detection in aerial images captured during wildlife surveys. *AI 2015: Advances in Artificial Intelligence*, pages 379–385, 2015.
- [2] Amanda Hodgson, Natalie Kelly, and David Peel. Unmanned aerial vehicles (uavs) for surveying marine fauna: a dugong case study. *PLoS ONE*, 8(11):e79556, 2013.
- [3] Patrick C. Gray, Abram B. Fleishman, David J. Klein, et al. A convolutional neural network for detecting sea turtles in drone imagery. *Methods in Ecology and Evolution*, 10(3):345–355, 2019.
- [4] Justine Boulent et al. Automatic detection of cetaceans in aerial images: challenges and perspectives. *Remote Sensing in Ecology and Conservation*, 2023.
- [5] Colleen M. Corrigan et al. Distribution of the humpback whale and other cetacean species from aerial surveys. *Marine Mammal Science*, 2014.
- [6] Ben Bartlett, Matheus Santos, Petar Trslc, and Gerard Dooly. Supporting offshore wind growth: Automating data analysis in digital aerial surveys to enhance wildlife protection and survey efficiency. *Ecological Informatics*, 90:103242, 2025. doi: 10.1016/j.ecoinf.2025.103242.
- [7] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 779–788, 2016.
- [8] Glenn Jocher, Jing Qiu, and Ayush Chaurasia. Ultralytics yolo. <https://github.com/ultralytics/ultralytics>, 2023. Version 8.x; accessed 2026.
- [9] Rahima Khanam and Muhammad Hussain. Yolov11: An overview of the key architectural enhancements. *arXiv preprint arXiv:2410.17725*, 2024.
- [10] Fatih Cagatay Akyon, Sinan Onur Altinuc, and Alptekin Temizel. Slicing aided hyper inference and fine-tuning for small object detection. *IEEE International Conference on Image Processing (ICIP)*, pages 966–970, 2022.
- [11] Nir Aharon, Roy Orfaig, and Ben-Zion Bobrovsky. Bot-sort: Robust associations multi-pedestrian tracking. In *arXiv preprint arXiv:2206.14651*, 2022.

- [12] Philip S. Hammond, Kelly Macleod, Per Berggren, David L. Borchers, Louise Burt, Ana Cañadas, Géneviève Desportes, Greg P. Donovan, Anita Gilles, Douglas Gillespie, Jonathan Gordon, Lex Hiby, Iwona Kuklik, Russell Leaper, Kristina Lehnert, Mardik Leopold, Phil Lovell, Nils Øien, Charles G. M. Paxton, Vincent Ridoux, Emer Rogan, Filipa Samarra, Meike Scheidat, Miguel Sequeira, Ursula Siebert, Henrik Skov, Rene Swift, Mark L. Tasker, Jonas Teilmann, Olivier Van Canneyt, and Jose Antonio Vázquez. Cetacean abundance and distribution in european atlantic shelf waters to inform conservation and management. *Biological Conservation*, 164:107–122, 2013. doi: 10.1016/j.biocon.2013.04.010.
- [13] Maxim Tkachenko, Mikhail Malyuk, Andrey Holmanyuk, and Nikolai Liubimov. Label studio: Data labeling software. <https://github.com/HumanSignal/label-studio>, 2020–2024.

A Reproducibility and Artefacts

- **Detector:** Ultralytics YOLO11 (n / s), single class, input 1024 px, fixed seed 1337.
- **Curation:** a dataset of 27,290 images / 55,163 boxes / 640 background frames over ten public sources (Table 2), after MD5 deduplication, EXIF-orientation correction and offline-augmentation collapse.
- **Screening (FROC):** two-stage SAHI predict-and-cache then recall@FPPI rescore on the held-out HiDef survey (704 frames); SAHI 0.12.1, tile 1024, overlap 0.2, IoU ≥ 0.3 .
- **Scale-adaptive inference:** the shared `cetacean.inference` package (geometry solver, scale cascade, SAHI runner), reused unchanged by the ROS 2 node of Section 7; sweep canvas 8192×5460 (DJI P1), APEM frame 14204×10652 (Phase One iXM-RS150F, EXIF altitude 448.5 m).
- **Pseudo-labelling:** human-in-the-loop review in Label Studio; the reviewed `gommaps` set = 420 frames / 1,874 boxes.
- **Hardware:** training and inference on a single Quadro RTX 4000 (8 GB); deployment target NVIDIA Jetson Orin class with TensorRT export.